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DEPARTMENT OF INFORMATION TECHNOLOGY

REVIEW 1

PRIVACY-PRESERVING INTRUSION DETECTION USING FEDERATED LEARNING FOR DISTRIBUTED IoT AND VEHICULAR NETWORKS

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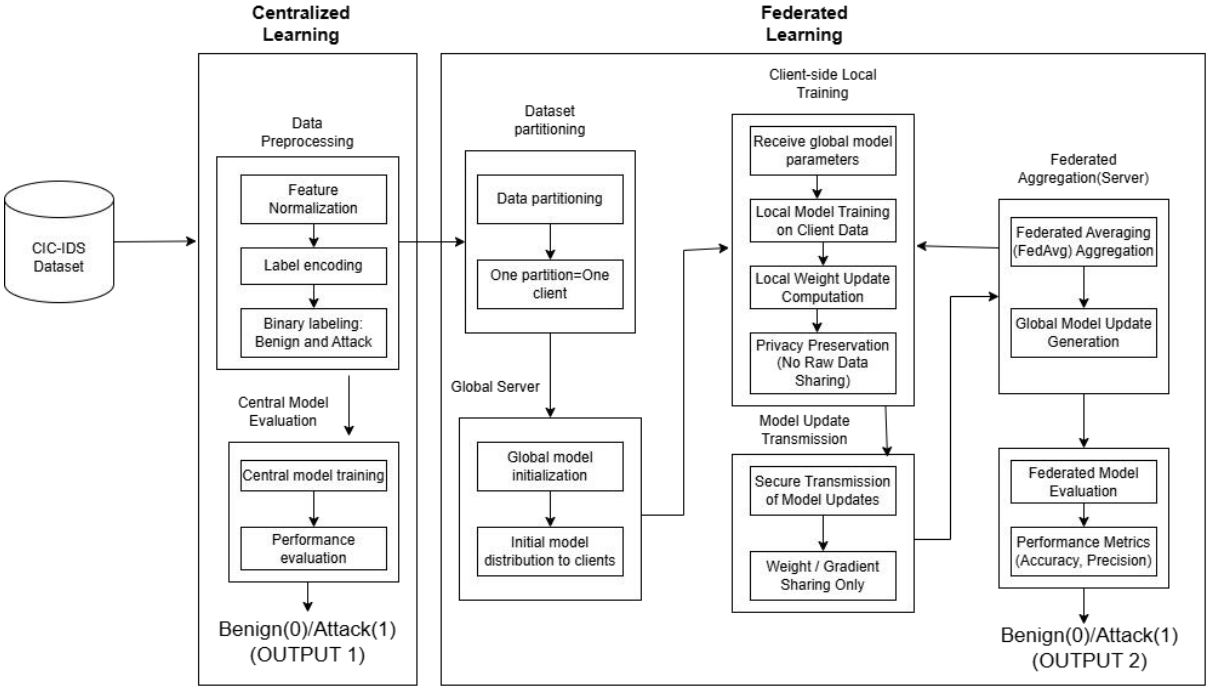
ARCHITECTURE DIAGRAM



1. ABSTRACT

The rapid growth of Internet of Things (IoT) and vehicular networks has significantly increased the attack surface for cyber intrusions. Traditional centralized intrusion detection systems (IDS) require aggregation of large volumes of network traffic data, leading to serious privacy, security, and scalability concerns. This project proposes a privacy-preserving intrusion detection framework using Federated Learning (FL), where multiple distributed clients collaboratively train a global IDS model without sharing raw data. Using the CIC-IDS2018 dataset, a binary classification system is developed to distinguish benign and malicious traffic.

2. INTRODUCTION

Vehicular IoT and distributed IoT environments generate massive volumes of network traffic that are vulnerable to attacks such as Denial of Service (DoS), Distributed DoS (DDoS), brute-force attacks, and botnets. Conventional centralized intrusion detection approaches require transferring sensitive network data to a central server, raising privacy risks and communications-overhead.

To address these challenges, Federated Learning enables decentralized model training where only model parameters are shared among clients, and raw data remains local. This project explores the feasibility of applying federated learning to intrusion detection systems while maintaining competitive detection performance.

3. METHODOLOGY

3.1 DATASET DESCRIPTION

The project uses the CIC-IDS2018 dataset, a large-scale intrusion detection dataset containing realistic network traffic flows.
Key characteristics of the dataset include:

- Contains both benign traffic and multiple attack types such as DoS, DDoS, Botnet, Brute Force, and SQL Injection
- Each record represents a network flow with statistical features such as flow duration, packet counts, byte rates, and inter-arrival times
- Dataset size of approximately 2.8 GB, provided as multiple CSV files
- Used for binary classification: Benign (0) and Attack (1)

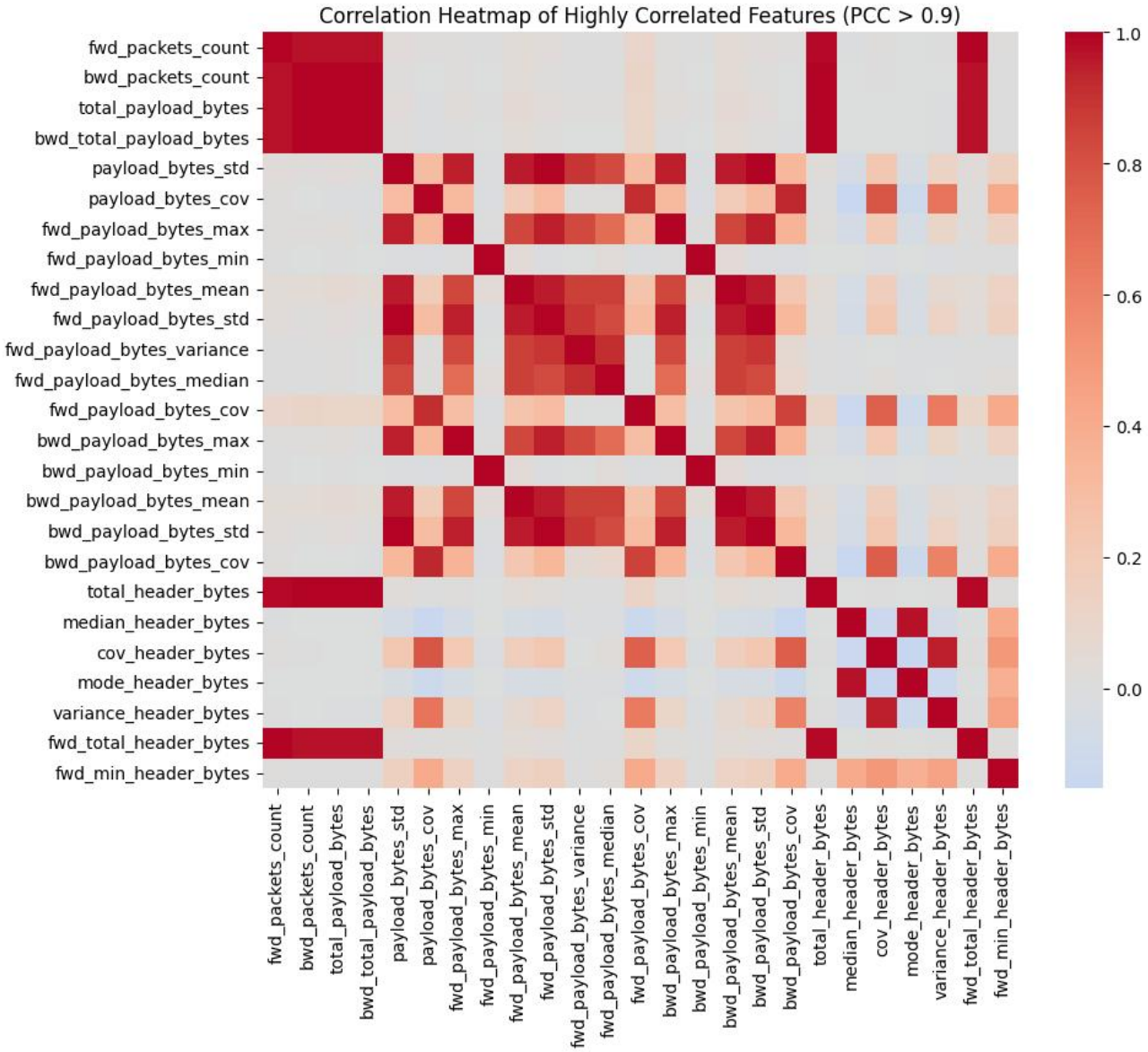
3.2 DATA PREPROCESSING

All CSV files corresponding to different attack scenarios are merged into a single dataset. The preprocessing steps include:

- Removal of identifier and irrelevant columns
- Handling missing values
- Replacing infinite values
- Assigning labels: Benign → 0, Attack → 1

Feature selection is performed using the Pearson Correlation Coefficient (PCC) to remove highly correlated features. After feature reduction, the final dataset contains 156 important numerical features from more than 310 features, reducing the payload by half, thus improving the performance during training.

Feature Reduction by Correlation Heatmap:



Feature Reduction Using PCC:

```
PCC sample shape: (39200, 315)
Total original features: 315
Features dropped by PCC: 159
Remaining features after PCC: 156

Sample dropped features: ['fwd_packets_count', 'bwd_packets_count', 'total_payload_bytes', 'bwd_total_payload_bytes', 'payload_bytes_std', 'paylo
ad_bytes_cov', 'fwd_payload_bytes_max', 'fwd_payload_bytes_min', 'fwd_payload_bytes_mean', 'fwd_payload_bytes_std']

Sample kept features: ['src_port', 'dst_port', 'duration', 'packets_count', 'fwd_total_payload_bytes', 'payload_bytes_max', 'payload_bytes_min',
'payload_bytes_mean', 'payload_bytes_variance', 'payload_bytes_median']
```

3.3 MODEL TRAINING

Centralized Learning

In centralized learning, the entire preprocessed dataset is available at a single node. Logistic Regression is used as the baseline model to establish an upper-bound performance benchmark for intrusion detection.

Federated Learning

For the federated setup, the dataset is partitioned into five simulated clients with an IID data distribution. Each client performs local model training, and the central server aggregates the model updates using FedAvg. Only model parameters are exchanged, ensuring privacy preservation.

Client Partitioning:

```
Client 1
label
0.0    0.750153
1.0    0.249847
Name: proportion, dtype: float64

Client 2
label
0.0    0.751318
1.0    0.248682
Name: proportion, dtype: float64

Client 3
label
0.0    0.749943
1.0    0.250057
Name: proportion, dtype: float64

Client 4
label
0.0    0.750554
1.0    0.249446
Name: proportion, dtype: float64

Client 5
label
0.0    0.750834
1.0    0.249166
Name: proportion, dtype: float64
```

Both centralized and federated models are trained for binary intrusion detection.

4. LOGISTIC REGRESSION MODEL

The Logistic Regression model operates as follows:

- Input consists of PCC-selected and normalized network traffic features
- A linear combination of input features is computed
- A sigmoid function maps the output to a probability between 0 and 1
- Based on a predefined threshold, traffic is classified as Benign or Attack
- The model is trained by minimizing log-loss using gradient-based optimization

5. RESULTS AND ANALYSIS

5.1 CENTRALIZED LEARNING RESULTS

- Dataset: Preprocessed CIC-IDS2018
- Training Mode: Centralized
- Models: Logistic Regression, CNN (PCC-CNN inspired)
- Accuracy achieved: ~85%

Centralized learning usually achieves the highest accuracy since the model is trained on the complete dataset. However, it requires full data sharing, which raises privacy and scalability concerns in vehicular IoT environments. That's why this Federated Learning method is being used, where the Data Sharing is never done and thus the privacy is prevented and only the updated Model Weights are updated to the central model and thus the data Privacy is preserved.

5.2 FEDERATED LEARNING RESULTS

- Dataset: Same preprocessed CIC-IDS2018
- Number of clients: 5
- Aggregation algorithm: FedAvg
- Training Mode: Decentralized
- Accuracy achieved: 94–97%

Although there is a slight drop in accuracy compared to centralized learning due to data partitioning and aggregation effects, the federated model maintains high detection performance while ensuring data privacy.

FED-Learning Training Phase(5 epochs & 5 clinets for each epoch):

```
================ FL ROUND 1 ================
Client 1 local accuracy: 0.8972
Client 2 local accuracy: 0.8976
Client 3 local accuracy: 0.8978
Client 4 local accuracy: 0.8974
Client 5 local accuracy: 0.8975
--> Server aggregated accuracy (Round 1): 0.9400

================ FL ROUND 2 ================
Client 1 local accuracy: 0.8971
Client 2 local accuracy: 0.8976
Client 3 local accuracy: 0.8978
Client 4 local accuracy: 0.8975
Client 5 local accuracy: 0.8975
--> Server aggregated accuracy (Round 2): 0.9409

================ FL ROUND 3 ================
Client 1 local accuracy: 0.8974
Client 2 local accuracy: 0.8979
Client 3 local accuracy: 0.8985
Client 4 local accuracy: 0.8978
Client 5 local accuracy: 0.8982
--> Server aggregated accuracy (Round 3): 0.9416

================ FL ROUND 4 ================
Client 1 local accuracy: 0.8975
Client 2 local accuracy: 0.8982
Client 3 local accuracy: 0.8988
Client 4 local accuracy: 0.8980
Client 5 local accuracy: 0.8986
--> Server aggregated accuracy (Round 4): 0.9421

================ FL ROUND 5 ================
Client 1 local accuracy: 0.8983
Client 2 local accuracy: 0.8982
Client 3 local accuracy: 0.8992
Client 4 local accuracy: 0.8981
Client 5 local accuracy: 0.8992
--> Server aggregated accuracy (Round 5): 0.9429
```

Centralized Aggregation Term:

Server computes:

$$W_g^{(r+1)} = \frac{1}{5}\sum_{i=1}^{5} W_i^{(r)}$$

Model Comparison(Centralized VS FED-Learning):

Metric	**Centralized learning**	**Federated Learning**
Training	Single centralized server	Distributed across multiple clients
Data	Entire dataset at one node	Data partitioned across clients
Number of clients	Not applicable	5 clients
Classification type	Binary	Binary
Model types	Logistic regression	Logistic regression
Accuracy	85%	94%
Privacy	No	Yes
Raw data sharing	Required	Not shared
Suitability for vehicle IoT	Low	High

6. NOVELTY

- Introduces a Federated Learning-based IDS for vehicular IoT environments
- Ensures privacy-preserving intrusion detection by keeping raw data local
- Combines PCC-based feature selection with machine learning models
- Provides a direct comparison between centralized and federated learning
- Demonstrates that federated learning achieves high accuracy with minimal performance loss

7. CONCLUSION

This project demonstrates the effectiveness of federated learning for intrusion detection in distributed IoT and vehicular networks. By eliminating the need for centralized data collection, the proposed approach enhances privacy while maintaining strong detection performance. The experimental results confirm that federated learning is a viable and scalable solution for real-world vehicular IoT security applications.